

National Institute of Technology Kurukshetra
B. Tech, 2nd Sessional Examination, 2026

Branch	: R&A	Course Code	: RAPC-210
Semester	: 4 th	Time	: 50 minutes
Course Name	: Actuators Drives and Sensors	Max. Marks	: 15
Time	: 04:00-04:50 PM	Date	: 08/04/2026

NOTE: Write answers clearly and to the point. Draw neat sketches wherever required.

- Q1 In a multi-cylinder actuation system, signal overlapping and conflicting commands can lead to malfunctioning. Explain how the CASCADE method eliminates these issues compared to a normal actuator circuit. [4]
- Q2 An automated conveyor system is used to segregate parts based on their height, using pneumatic actuators. The system consists of: i) A conveyor belt carrying mixed-height components, ii) A height sensor (S1) to classify parts into two categories (low, and high), and iii) two double-acting pneumatic cylinders: Cylinder A → pushes low-height parts into bin 1, and Cylinder B → pushes high-height parts into bin 2. Based on this setup, draw the electro-pneumatic circuit diagram for the system using interlocking to ensure that only one cylinder actuates at a time. (Hint: use AND valve and Pilot operated valve) [6]
- Q3 Discuss the role of the following mechanical systems in automation. Support your answer with suitable examples: (attempt any three) [5]
- a) Gear
 - b) Cam
 - c) Rack and Pinion
 - d) Bearing
 - e) Belt Drive